

Morgan Grougan

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Education

May 2026 **B.S. in Biochemistry, Minor in Computer Science**, Wayne State University, Detroit, MI

Awards and Honors

Summer 2023 David and Beverly Rorabacher Research Award

Fall 2023 - Maximizing Access to Research Careers (MARC) Scholarship
Spring 2025

9/14/2024 Selected Speaker, RebuildDetroit Conference

Fall 2025 and Winter 2026 Dale H. Chidester Memorial Endowed Scholarship in Chemistry

Coding Projects

Librarian of Alexandria (LoA), <https://github.com/MorganRO8/LoA>

- A python based tool for automated scraping and subsequent extraction of structured scientific data from research papers
- Automates literature review process and generates large-scale data for analysis in scientific research

ChromoGen, <https://github.com/MorganRO8/ChromoGen>

- A novel python based reinforcement-learning pipeline which uses random forest based chemical property prediction models to guide molecule generation
- XGBoost random-forest model was trained on data assembled using LoA

RetroCluster, <https://github.com/MorganRO8/RetroCluster>

- A novel clustering method for chemical reactions which uses a tiered and weighted system of reaction conditions, mechanism, and structures to organize chemical reactions
- Part of a larger project to build a retrosynthesis planning machine capable of producing realistic synthesis pathways for novel targets

CryoEM, <https://github.com/MorganRO8/CryoML>

- Python/C++ CUDA kernel (via CuPy) code designed for a Kaggle competition to detect the locations of proteins within Cryogenic Electron Microscopy images
- Created custom 3D watershed algorithm for object detection within imaged spaces before passing to a number of different types of machine learning protein identification models

Lucy's Labyrinth, https://github.com/MorganRO8/Lucys_Labyrinth

- A video game coded from scratch in C++ as a class project
- Implemented Mistral-7B based prompt system in video game for flexible dialogue and to avoid hard-coded choices

Publications

Morgan Grougan, Mark A. Hix, and Alice R. Walker. **Librarian of Alexandria: An extensible LLM-based chemical data extraction pipeline. Available on ACS**, <https://pubs.acs.org/doi/10.1021/acs.jcim.6c00374>

Sebastien Hebert, Janya Subasinghe, Morgan Grougan, and Alice R. Walker.
Effect of oligomerization on fluorescent protein DsRed excited state dynamics and brightness. In preparation.

Research Experience

- Dec 2022–Present **Undergraduate Research Assistant**, *Adviser: Alice R. Walker, Department of Chemistry, Wayne State University*
- Prepared PDB files for QM/MD simulations in physiological conditions
 - Prepared and ran TeraChem job files for simulations of proteins and small molecules on a distributed high-performance computing cluster
 - Developed a novel large language model based pipeline to automate the process of finding and extracting data from chemical literature
 - Designed novel clustering approach to analyze hydrogen bonding networks of monomeric pieces of oligomeric proteins in QM/MD simulations to classify functional differences between components with identical amino acid sequences
 - Designed a novel 3D watershed algorithm and implemented it as code in a custom CuPy kernel for rapid detection of protein candidates in Cryogenic Electron-Microscopy images
 - Trained XGBoost models to predict the properties of molecules within their localized region of chemical space
 - Developed a means by which the property prediction models I trained can be used to guide reinforcement learning and novel molecule generation
 - Developed a custom clustering algorithm for chemical reactions to address the issues with current methods for clustering reactions to produce reaction groups which align more closely with human defined reaction types

Talks and Posters

- Fall 2023 Wayne State College of Liberal Arts and Sciences Undergraduate Symposium*
- April 2024 Maximizing Access to Research Careers End-of-Year Symposium
- November 2024 ReBUILDetroit Selected Speaker - 15 Minute Talk
- April 2025 Maximizing Access to Research Careers End-of-Year Symposium

Skills

- Computational Chemistry Geometry Optimization, Molecular Dynamics and Quantum Mechanics Based Simulations, Simulation of Biological Environments, Small Molecule and Protein Simulation
- Machine Learning Property prediction, reinforcement learning, drug design & candidate screening, LLM design and implementation
- Programming Python (Pandas, pytorch, SKLearn, XGBoost, etc.), C++, parallelization, GPU acceleration (CUDA)
- Other Data analysis and visualization, Database Management

Professional Development/Teaching

- MARC Scholarship Program Learning Community, Weekly sessions
- Weekly machine learning sessions, both attending and leading/teaching

Languages

- English Native